Experiment 8

AIM: Use Postman to test sample API

1. Objective

- Understand how to test REST APIs using Postman by sending different types of HTTP requests.
  - Get
  - Post
  - Put
  - Delete

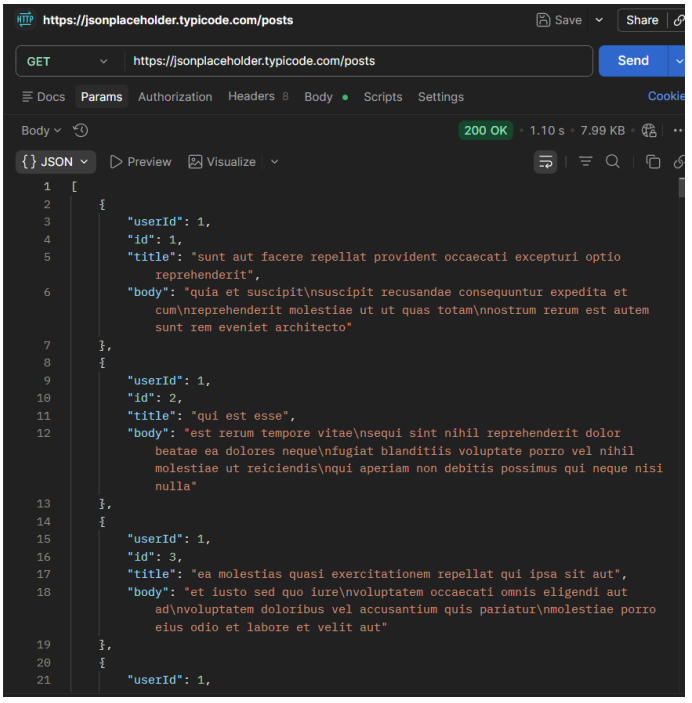
2. Prerequisites

- postman
- API to be tested
  - Create a simple node.js API / use some free fake API

3. Test API

Get:

URL: https://jsonplaceholder.typicode.com/posts



Post:

https://jsonplaceholder.typicode.com/posts

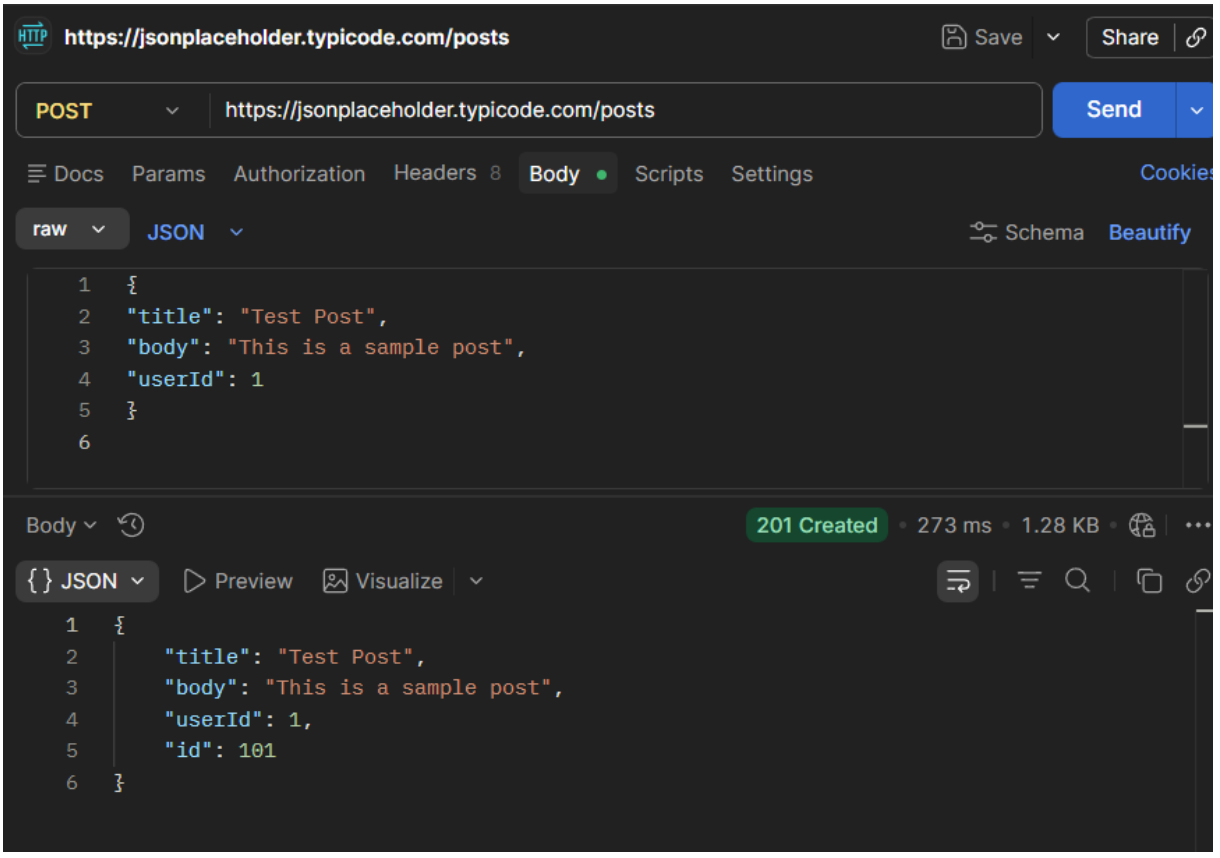
{

"title": "Test Post",

"body": "This is a sample post",

"userId": 1

}



Put:

https://jsonplaceholder.typicode.com/posts/1

{

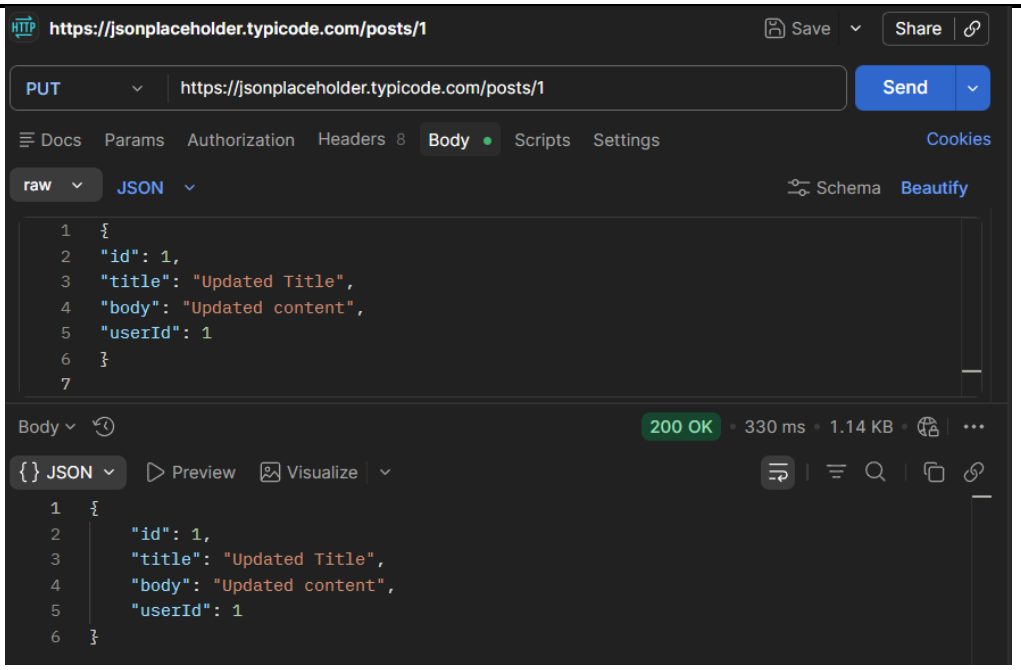
"id": 1,

"title": "Updated Title",

"body": "Updated content",

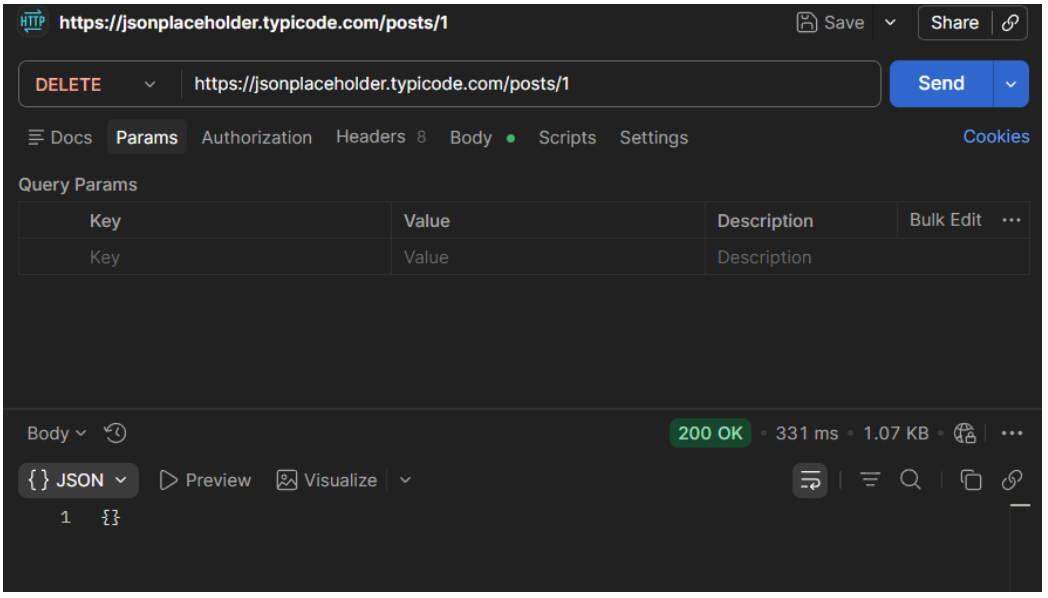
"userId": 1

}



Delete:

https://jsonplaceholder.typicode.com/posts/1



4. Create node.js API

Step 1: Create Project

mkdir simple-api

cd simple-api

npm init -y

npm install express

Step 2: Create server.js file

```
const express = require('express');
const app = express();
app.use(express.json());
let students = [
{ id: 1, name: "Viraj", branch: "CE" },
{ id: 2, name: "Ravi", branch: "CE" }
];
app.get('/students', (req, res) => {
res.json(students);
});
app.get('/students/:id', (req, res) => {
const student = students.find(s => s.id == req.params.id);
res.json(student);
});
app.post('/students', (req, res) => {
const newStudent = {
id: students.length + 1,
name: req.body.name,
branch: req.body.branch
};
students.push(newStudent);
res.status(201).json(newStudent);
});
app.put('/students/:id', (req, res) => {
const student = students.find(s => s.id == req.params.id);
if (student) {
student.name = req.body.name;
student.branch = req.body.branch;
res.json(student);
} else {
```

```
res.status(404).json({ message: "Student not found" });

}

});

// DELETE - Remove student

app.delete('/students/:id', (req, res) => {

students = students.filter(s => s.id != req.params.id);

res.json({ message: "Student deleted" });

});

app.listen(3000, () => {

console.log("Server running on http://localhost:3000");

});
```

Step 3: Run server

Node server.js

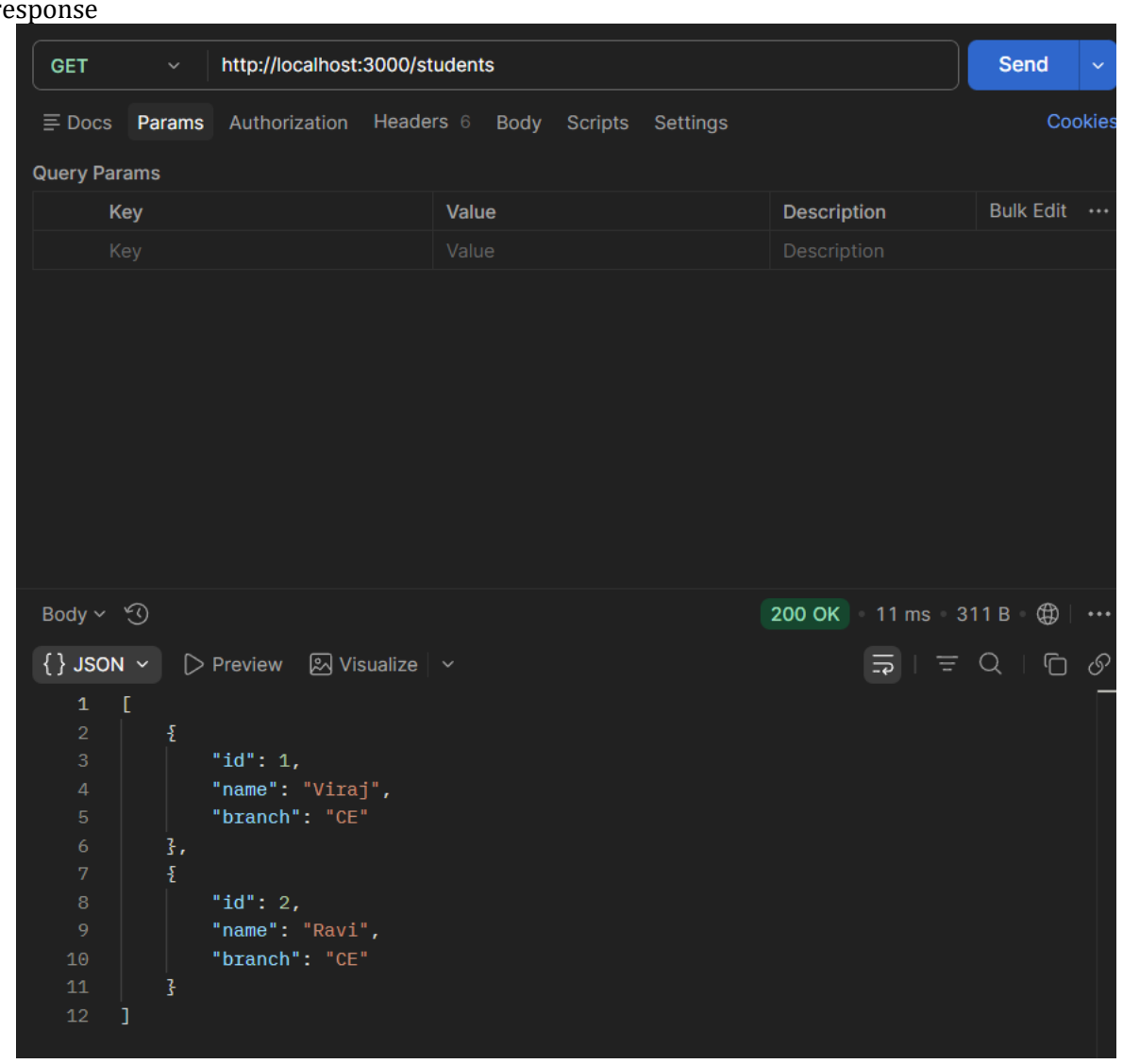
Command :

```
C:\Users\RAJNISH\Desktop\OLD DATA\B.Tech\OST>mkdir simple-api

C:\Users\RAJNISH\Desktop\OLD DATA\B.Tech\OST>cd simple-api

C:\Users\RAJNISH\Desktop\OLD DATA\B.Tech\OST\simple-api>npm init -y
Wrote to C:\Users\RAJNISH\Desktop\OLD DATA\B.Tech\OST\simple-api\package.json:

{
  "name": "simple-api",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}

C:\Users\RAJNISH\Desktop\OLD DATA\B.Tech\OST\simple-api>npm install express

added 65 packages, and audited 66 packages in 5s

22 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

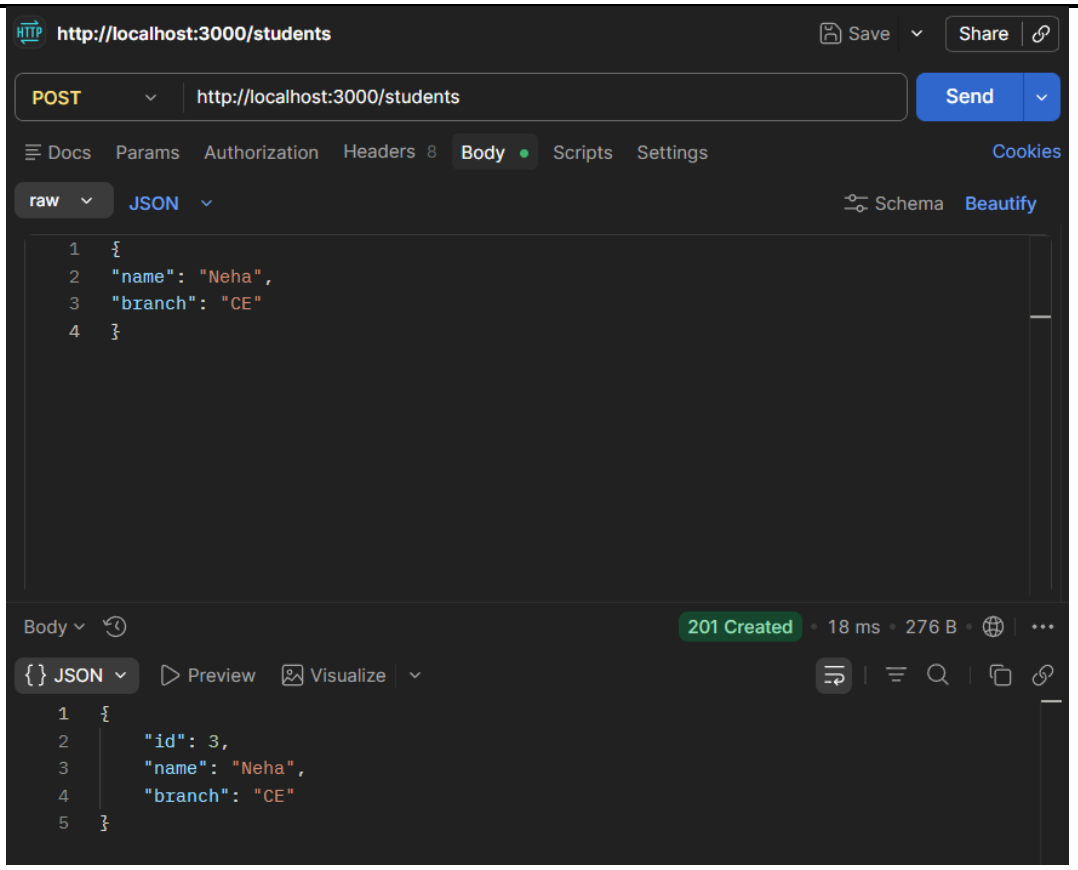
C:\Users\RAJNISH\Desktop\OLD DATA\B.Tech\OST\simple-api>notepad server.js

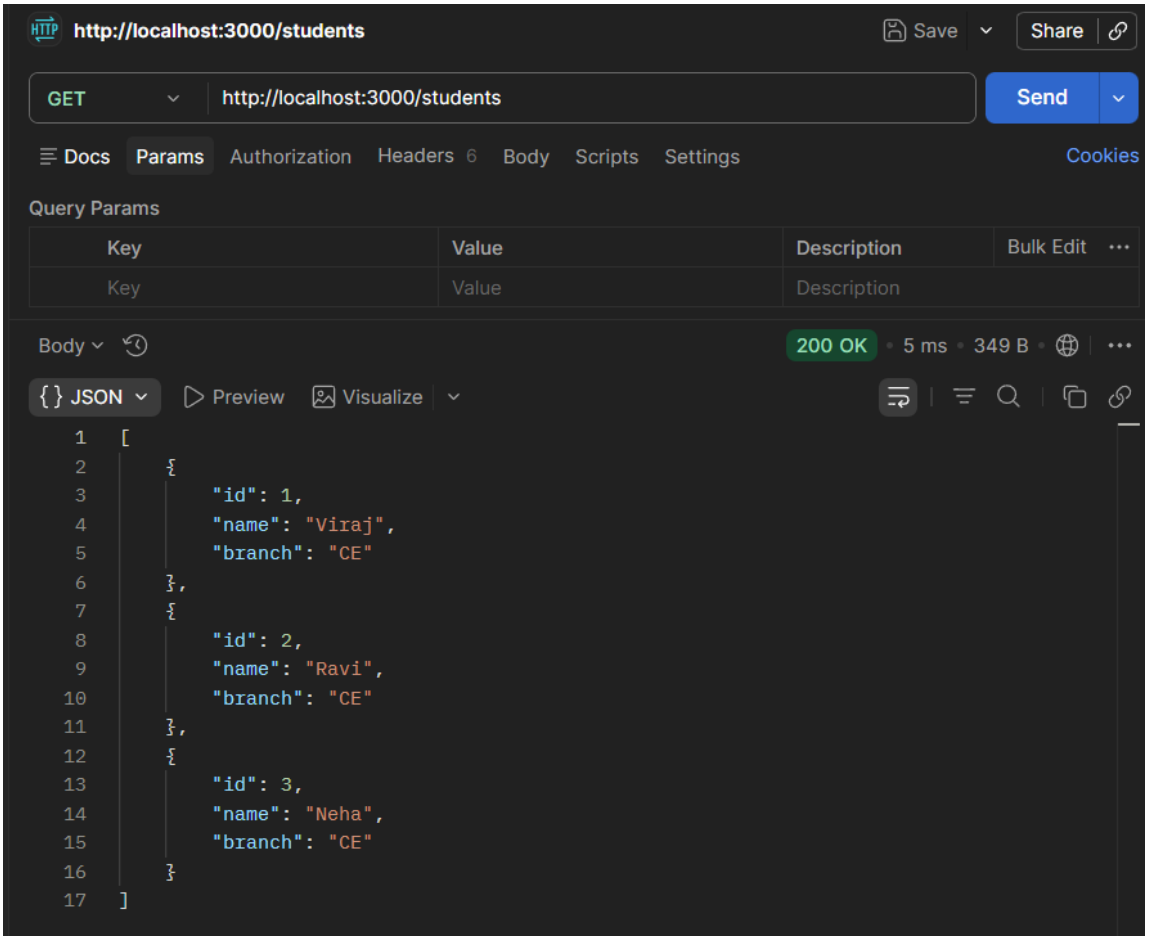
C:\Users\RAJNISH\Desktop\OLD DATA\B.Tech\OST\simple-api>node server.js
Server running on http://localhost:3000

5. Procedure TEST using postman
1. GetAll:
Type: Get
URL : http://localhost:3000/students
Check response

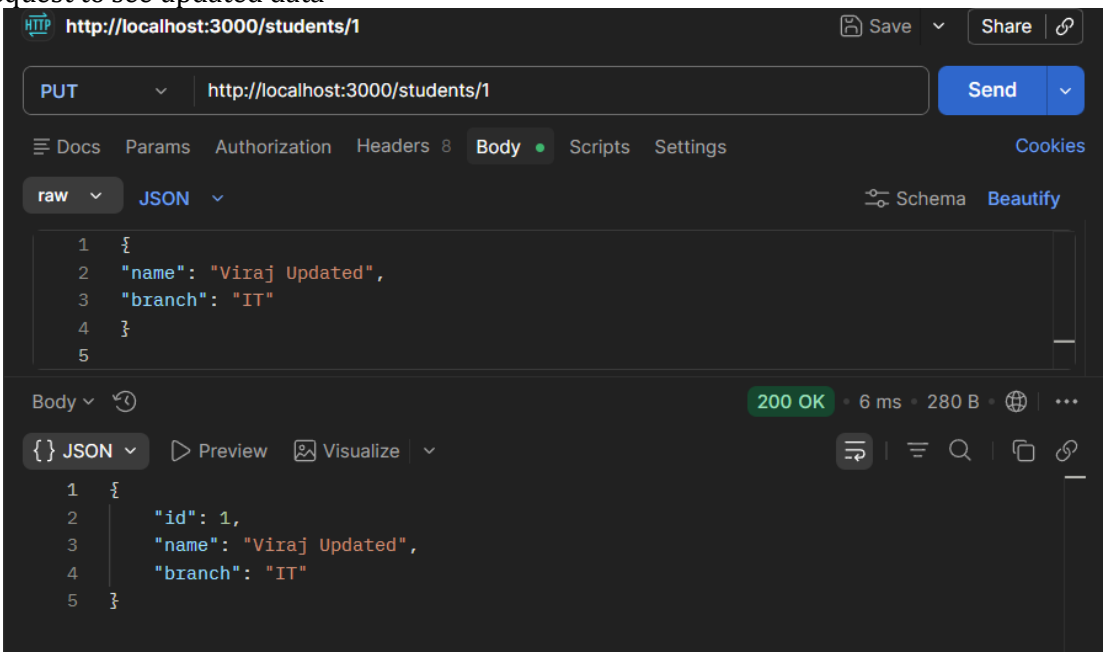


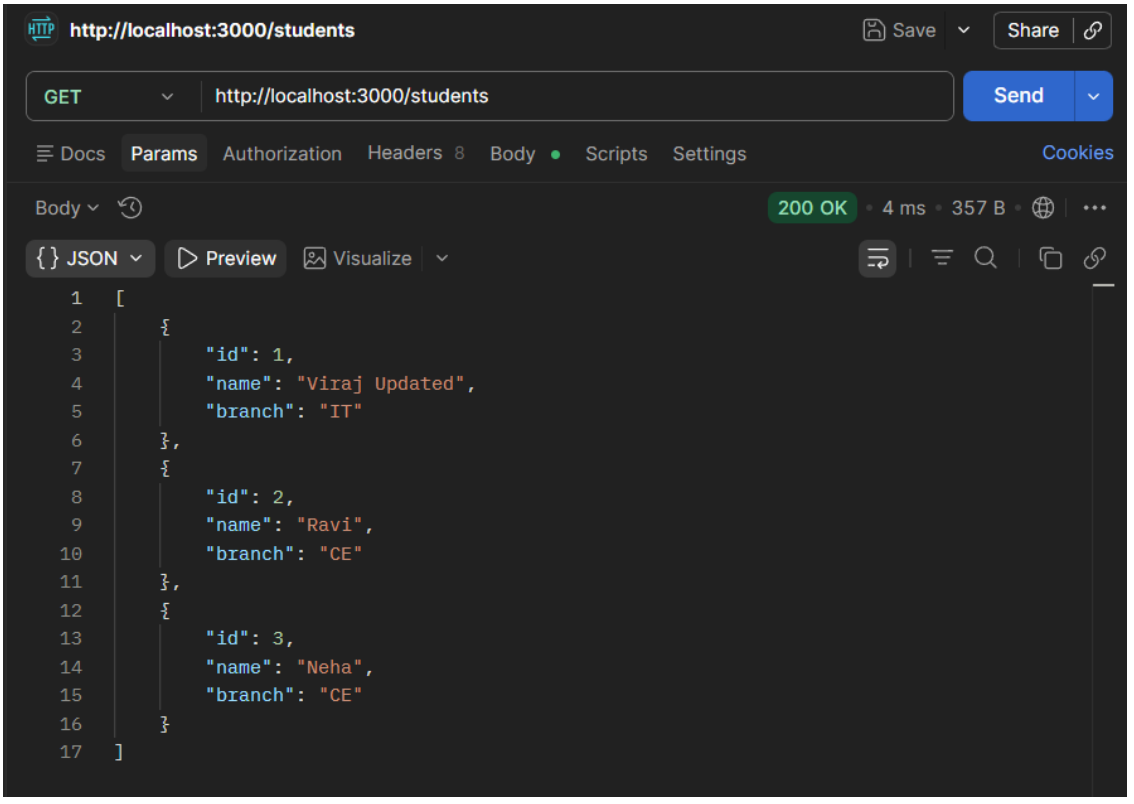
2. Post:
Type: Post
URL: http://localhost:3000/students
Body:
{
"name": "Neha",
"branch": "CE"
}
Check response and Send Get request to see new data inserted


http://localhost:3000/students
Save
Share
POST
http://localhost:3000/students
Send
Docs
Params
Authorization
Headers 8
Body
Scripts
Settings
Cookies
raw
JSON
Schema
Beautify
{
"name": "Neha",
"branch": "CE"
}
Body
201 Created
18 ms
276 B
JSON
Preview
Visualize
{
"id": 3,
"name": "Neha",
"branch": "CE"
}


http://localhost:3000/students
Save
Share
GET
http://localhost:3000/students
Send
Docs
Params
Authorization
Headers 6
Body
Scripts
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
Key
Value
Description
Body
200 OK
5 ms
349 B
JSON
Preview
Visualize
[
{
"id": 1,
"name": "Viraj",
"branch": "CE"
},
{
"id": 2,
"name": "Ravi",
"branch": "CE"
},
{
"id": 3,
"name": "Neha",
"branch": "CE"
}
]

3. Put request:
Type: PUT
URL : http://localhost:3000/students/1
Body:
{
"name": "Viraj Updated",
"branch": "IT"
}
Check response
Send Get request to see updated data

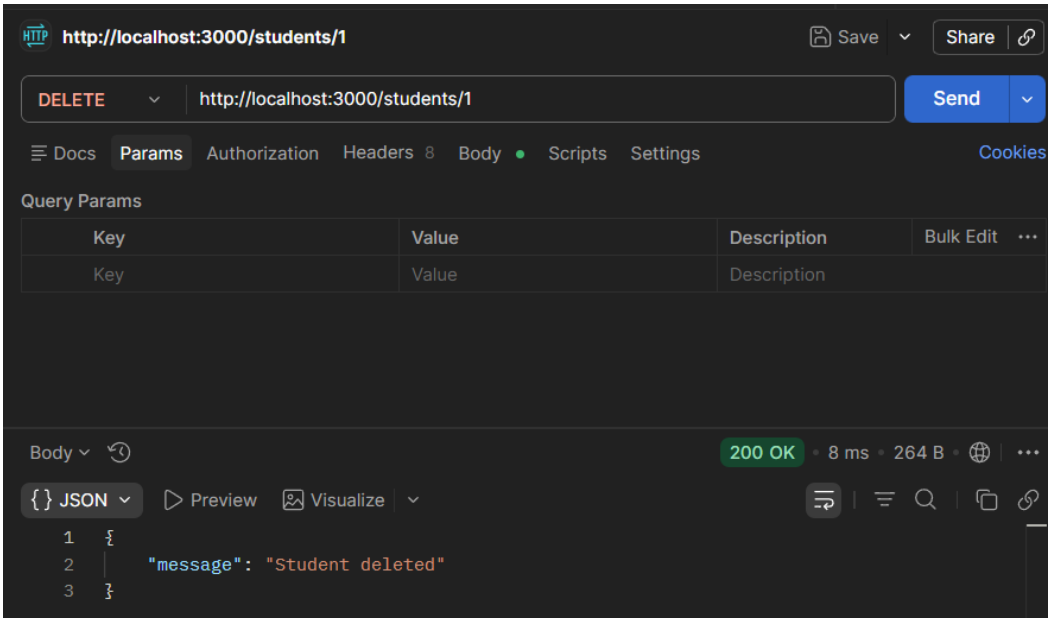
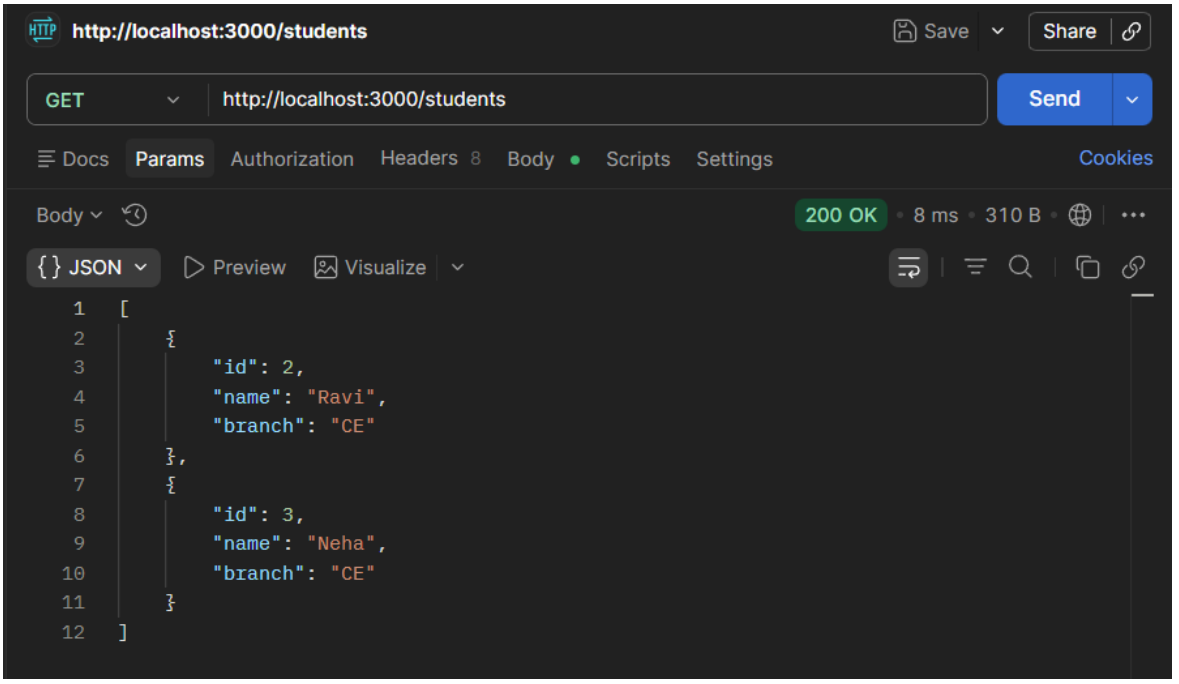




4. Delete request:

Type: DELETE

URL : http://localhost:3000/students/1

Check response

Send Get request to see updated data